Numerical Methods-II

Partial differential equations occur in many branches of applied mathematics, for example, in hydrodynamics, elasticity, quantum mechanics and electromagnetic theory. The analytical treatment of these equations is a rather involved process and requires application of advanced mathematical methods. On the other hand, it is generally easier to produce sufficiently approximate solutions by simple and efficient numerical methods. Several numerical methods have been proposed for the solution of partial differential equations, but only finite difference methods have become popular and are more gainfully employed than others.

Finite difference approximations to partial derivatives.

By the Taylor's series, we have

$$u(x+h, y) = u(x,y) + h u_x(x,y) + \frac{1}{2} h^2 u_{xx}(x,y) + \frac{h^3}{3!} u_{xxx}(x,y) + \cdots \rightarrow (1)$$

$$u(x-h, y) = u(x,y) - h u_x(x,y) + \frac{h^2}{2} u_{xx}(x,y) - \frac{h^3}{3!} u_{xxx}(x,y) + \cdots \rightarrow (2)$$

$$u(x, y+k) = u(x,y) + k u_y(x,y) + \frac{k^2}{2} u_{yy}(x,y) + \frac{k^3}{3!} u_{yyy}(x,y) + \rightarrow (3)$$

$$u(x, y-k) = u(x,y) - k u_y(x,y) + \frac{k^2}{2} u_{yy}(x,y) - \frac{k^3}{3!} u_{yyy}(x,y) + \cdots \rightarrow (4)$$

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Finite difference approximations for partial derivatives in x-direction

① $\Rightarrow$ $u_x = \dfrac{u(x+h, y) - u(x, y)}{h}$ (Forward difference)

Neglecting $h^2, h^3, \ldots$

② $\Rightarrow$ $u_x = \dfrac{u(x, y) - u(x-h, y)}{h}$ (Backward difference)

① − ② $\Rightarrow$ $u_x = \dfrac{u(x+h, y) - u(x-h, y)}{2h}$ (Central difference)

For the second order partial derivative, we take

① + ②, by neglecting $h^3, h^4, \ldots$

$$u_{xx} = \frac{u(x+h, y) - 2u(x, y) + u(x-h, y)}{h^2}$$

Similarly finite difference approximations for partial derivatives in y-direction is

③ $\Rightarrow$ $u_y = \dfrac{u(x, y+K) - u(x, y)}{K}$ (Forward difference)

④ $\Rightarrow$ $u_y = \dfrac{u(x, y) - u(x, y-K)}{K}$ (Backward difference)

③ − ④ $\Rightarrow$ $u_y = \dfrac{u(x, y+K) - u(x, y-K)}{2K}$ (Central difference)

For the second order partial derivatives we have

③ + ④, by neglecting $K^3, K^4, \ldots$

$$u_{yy} = \frac{u(x, y+K) - 2u(x, y) + u(x, y-K)}{K^2}$$

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We can now obtain the finite difference analogues of partial differential equations by replacing the derivatives in any equation by their corresponding difference approximations.

Thus the Laplace equation in two dimensions, namely

$u_{xx} + u_{yy} = 0$ or $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ has its finite difference analogue

$$\frac{u(x+h,y) - 2u(x,y) + u(x-h,y)}{h^2} + \frac{u(x,y+k) - 2u(x,y) + u(x,y-k)}{k^2} = 0$$

If $h=k$, this gives

$$u(x+h,y) + u(x-h,y) + u(x,y+h) + u(x,y-h) = 4u(x,y)$$

where h is called the mesh size.

If we adopt the notation

$u(x,y) = u_{i,j}$, $u(x+h,y) = u_{i+1,j}$, $u(x-h,y) = u_{i-1,j}$

$u(x,y+h) = u_{i,j+1}$, $u(x,y-h) = u_{i,j-1}$ the above equation becomes

$$u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} = 4u_{i,j}$$

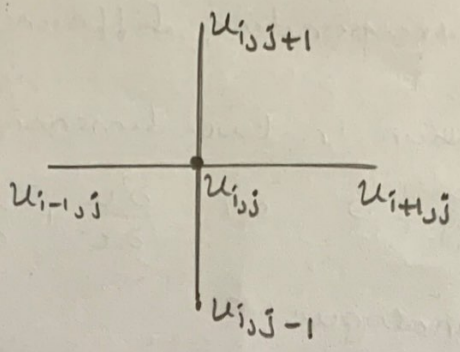
$$\Rightarrow \boxed{u_{i,j} = \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}}{4}} \rightarrow (1)$$

This shows that the value of $u_{i,j}$ at any interior mesh point is the average of its value at four neighbouring points to the left, right, above and below

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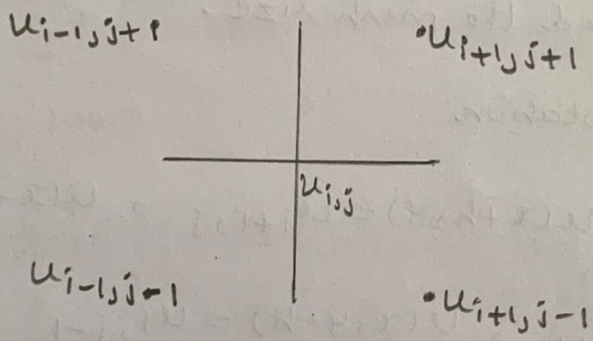
$u_1 = 781.25, u_2 = 1031.25$

Equation (1) is called the standard five point formula.



Diagonal five point formula is given by

$$u_{i,j} = \frac{u_{i-1,j+1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j-1}}{4}$$

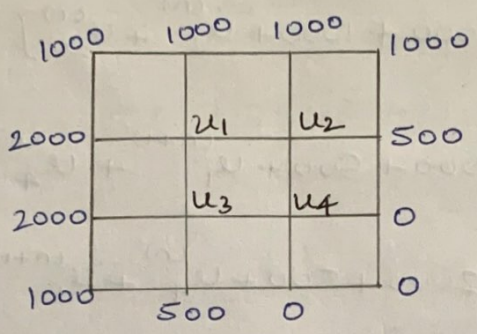


The error in the diagonal formula is four times that in the standard five point formula. Hence, in all computations we should prefer to use the standard five point formula wherever possible.

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Third iteration $n=2$ $(u_1 = 1187.5,\ u_2 = 781.25,\ u_3 = 1031.25$

① Given the values of $u(x,y)$ on the boundary of the square region. Evaluate the function $u(x,y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points of the figure.



Solve using Liebmann's method. Perform five iterations

Soln: To get the initial values of u_1, u_2, u_3 and u_4 we assume $u_4 = 0$.

Then $u_1 = \frac{1}{4}[1000 + 2000 + 1000 + u_4]$ (Diagonal five point formula)

$= \frac{1}{4}[1000 + 2000 + 1000 + 0]$ $(\because u_4 = 0)$

$= 1000$

$u_2 = \frac{1}{4}[1000 + u_1 + u_4 + 500]$ (Standard five point formula)

$= \frac{1}{4}[1000 + 1000 + 0 + 500]$

$= 625$

$u_3 = \frac{1}{4}[u_1 + 2000 + 500 + u_4]$ (Standard five point formula)

$= \frac{1}{4}[1000 + 2000 + 500 + 0]$

$= 875$

$u_4 = \frac{1}{4}[u_2 + u_3 + 0 + 0]$ (Standard five point formula)

$= \frac{1}{4}[625 + 875]$

$= 375$

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∴ initial approximation is $u_1 = 1000$, $u_2 = 625$, $u_3 = 875$,

$u_4 = 375$.

Using standard five point formula, we get

$$u_1^{(n+1)} = \frac{1}{4}\left[2000 + 1000 + u_3^{(n)} + u_2^{(n)}\right]$$

$$u_2^{(n+1)} = \frac{1}{4}\left[1000 + 500 + u_1^{(n+1)} + u_4^{(n)}\right]$$

$$u_3^{(n+1)} = \frac{1}{4}\left[2000 + 500 + u_4^{(n)} + u_1^{(n+1)}\right]$$

$$u_4^{(n+1)} = \frac{1}{4}\left[0 + 0 + u_3^{(n+1)} + u_2^{(n+1)}\right]$$

By Liebmann's iterative scheme, we get

First iteration, $n = 0$,

$$u_1^{(1)} = \frac{1}{4}[2000 + 1000 + 875 + 625] = 1125$$

$$u_2^{(1)} = \frac{1}{4}[1000 + 500 + 1125 + 375] = 750$$ (New value of u_1 is used)

$$u_3^{(1)} = \frac{1}{4}[2000 + 500 + 375 + 1125] = 1000$$

$$u_4^{(1)} = \frac{1}{4}[1000 + 750] = 437.5$$

Now, $u_1 = 1125$, $u_2 = 750$, $u_3 = 1000$, $u_4 = 437.5$

Second iteration $(n = 1)$

$$u_1^{(2)} = \frac{1}{4}[2000 + 1000 + 1000 + 750] = 1187.5$$

$$u_2^{(2)} = \frac{1}{4}[1000 + 500 + 1187.5 + 437.5] = 781.25$$

$$u_3^{(2)} = \frac{1}{4}[2000 + 500 + 437.5 + 1187.5] = 1031.25$$

$$u_4^{(2)} = \frac{1}{4}[1031.25 + 781.25] = 453.125$$

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Third iteration $n=2$ $(u_1=1187.5,\ u_2=781.25,\ u_3=1031.25,\ u_4=453.125)$

$$u_1^{(3)} = \frac{1}{4}[2000+1000+1031.25+781.25] = 1203.125$$

$$u_2^{(3)} = \frac{1}{4}[1000+500+1203.125+453.125] = 789.0625$$

$$u_3^{(3)} = \frac{1}{4}[2000+500+453.125+1203.125] = 1039.0625$$

$$u_4^{(3)} = \frac{1}{4}[1039.0625+789.0625] = 457.03125$$

Fourth iteration $(n=3)$ $(u_1=1203.125,\ u_2=789.0625,\ u_3=1039.0625,\ u_4=457.03125)$

$$u_1^{(4)} = \frac{1}{4}(2000+1000+1039.0625+789.0625) = 1207.03125$$

$$u_2^{(4)} = \frac{1}{4}(1000+500+1207.03125+457.03125) = 791.0156$$

$$u_3^{(4)} = \frac{1}{4}(2000+500+457.03125+1207.03125) = 1041.0156$$

$$u_4^{(4)} = \frac{1}{4}(1041.0156+791.0156) = 458.0078$$

Fifth iteration $(n=4)$ $(u_1=1207.0312,\ u_2=791.015,\ u_3=1041.01,\ u_4=458.0078)$

$$u_1^{(5)} = \frac{1}{4}(2000+1000+1041.015+791.015) = 1208.0078$$

$$u_2^{(5)} = \frac{1}{4}(1000+500+1208.0078+458.0078) = 791.5039$$

$$u_3^{(5)} = \frac{1}{4}(2000+500+458.0078+1208.0078) = 1041.5039$$

$$u_4^{(5)} = \frac{1}{4}(1041.5039+791.5039) = 458.2519$$

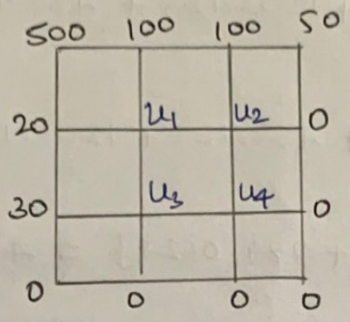
Hence the approximate values are

$u_1 = 1208.0078,\ u_2 = 791.5039,\ u_3 = 1041.50,\ u_4 = 458.2519$

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2) Solve $u_{xx}+u_{yy}=0$ in the following square region with the boundary conditions indicated as shown in the figure

Using Gauss-Seidel method.



Soln:- To get the initial values of u_1, u_2, u_3 and u_4 we assume $u_4=0$.

Then $u_1=\frac{1}{4}[500+100+30+u_4]$ (Diagonal five point formula)

$=\frac{1}{4}[500+100+30+0]$

$=157.5$

$u_2=\frac{1}{4}[100+u_1+u_4+0]$ (Standard five point formula)

$=\frac{1}{4}[100+157.5+0+0]$

$=64.375$

$u_3=\frac{1}{4}[u_1+30+0+u_4]$ (Standard five point formula)

$=\frac{1}{4}[157.5+30+0+0]$

$=46.875$

$u_4=\frac{1}{4}[u_2+u_3+0+0]$ (Standard five point formula)

$=\frac{1}{4}[64.375+46.875]$

$=27.8125$

Hence the ~~approx~~ initial approximation is

$u_1=157.5, u_2=64.375, u_3=46.875, u_4=27.8125$

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Using Standard five point formula, we get

$$u_1^{(n+1)} = \frac{1}{4}\left[100+20+u_2^{(n)}+u_3^{(n)}\right]$$

$$u_2^{(n+1)} = \frac{1}{4}\left[100+u_1^{(n+1)}+u_4^{(n)}+0\right]$$

$$u_3^{(n+1)} = \frac{1}{4}\left[u_1^{(n+1)}+30+0+u_4^{(n)}\right]$$

$$u_4^{(n+1)} = \frac{1}{4}\left[u_2^{(n+1)}+u_3^{(n+1)}+0+0\right]$$

Now using Gauss-seidel method with the known initial values, we get

iteration	u_1	u_2	u_3	u_4
1	57.8125	46.4062	28.9062	18.8281
2	48.8281	41.9140	24.4140	16.5820
3	46.5820	40.7910	23.2910	16.0205
4	46.0205	40.5102	23.0102	15.8801
5	45.8801	40.4400	22.9400	15.8450
6	45.8450	40.4225	22.9225	15.8362
7	45.8362	40.4181	22.9181	15.8340
8	45.8340	40.4170	22.9170	15.8335

Hence the approximate value of $u_1 = 45.8340$

$u_2 = 40.4170$

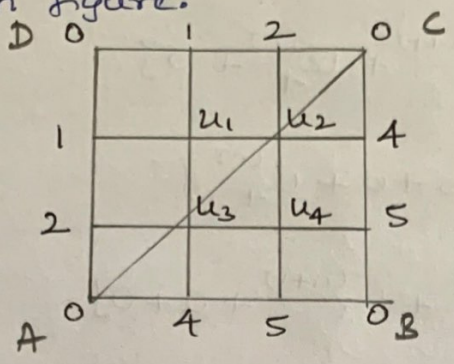
$u_3 = 22.9170$

$u_4 = 15.8335$

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Using standard five point formula, we get

3) Using Liebmann's method, solve the equation $u_{xx}+u_{yy}=0$ for the following square mesh with boundary values as shown in figure.



Soln:- The boundary values are symmetric with respect to line AC. Hence $u_2=u_3$.

∴ we have three unknowns u_1, u_2 and u_4.

To get initial values, we assume $u_2=0$ and $u_4=0$.

$$\therefore u_1 = \frac{1}{4}[1+1+u_3+u_2] \quad \text{(standard five point formula)}$$

$$= \frac{1}{4}[1+1+0+0] \quad (\because u_2=u_3)$$

$$= 0.5$$

$$u_2 = \frac{1}{4}[2+u_1+u_4+4] \quad \text{(standard five point formula)}$$

$$= \frac{1}{4}[2+0.5+0+4]$$

$$= 1.625$$

$$u_4 = \frac{1}{4}[u_2+u_3+5+5] \quad \text{(standard five point formula)}$$

$$= \frac{1}{4}[1.625+1.625+5+5]$$

$$= 3.3125$$

Hence the initial approximation is, $u_1=0.5$, $u_2=1.625$

$u_4 = 3.3125$

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Using Standard five point formula, we get

$$u_1^{(n+1)} = \frac{1}{4}(1+1+2u_2^{(n)}) \qquad (\because u_2 = u_3)$$

$$u_2^{(n+1)} = \frac{1}{4}(2+u_1^{(n+1)} + u_4^{(n)} + 4)$$

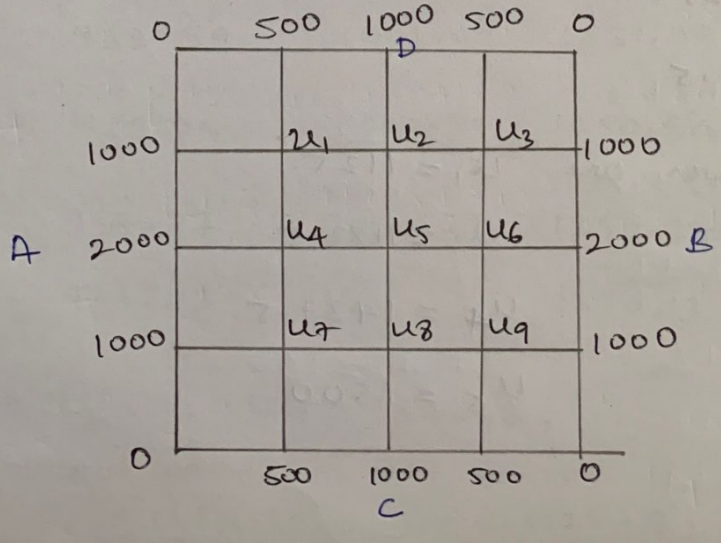
$$u_4^{(n+1)} = \frac{1}{4}(2u_2^{(n+1)} + 5+5) \qquad (\because u_2 = u_3)$$

∴ By Liebmann's method with initial values, we get

iteration	u_1	u_2	u_4
1	1.3125	2.6562	3.8281
2	1.8281	2.9140	3.9570
3	1.9570	2.9785	3.9892
4	1.9892	2.9946	3.9973
5	1.9973	2.9986	3.9993
6	1.9993	2.9996	3.9998

∴ Approximate values of $u_1 = 2$, $u_2 = 3$, $u_3 = 3$, $u_4 = 4$.

④ Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the square mesh with boundary values as shown using Liebmann's method.



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Soln:- Let $u_1, u_2, \ldots, u_9$ be the values of u at the interior mesh points. Since the boundary values of u are symmetrical about AB

$$u_1 = u_7, \; u_8 = u_2 \text{ and } u_3 = u_9$$

Also the values of u being symmetrical about line CD

$$u_3 = u_1, \; u_6 = u_7 \text{ and } u_9 = u_7$$

Thus it is sufficient to find the values u_1, u_2, u_4 and u_5

To find initial values, we use the following

$$u_5 = \frac{1}{4}(2000 + 2000 + 1000 + 1000) \quad \text{(Standard five point formula)}$$

$$u_5 = 1500$$

$$u_1 = \frac{1}{4}(0 + 1000 + u_5 + 2000) \quad \text{(Diagonal five point formula)}$$

$$u_1 = 1125$$

$$u_2 = \frac{1}{4}(1000 + u_1 + u_5 + 1000)$$

$$= \frac{1}{4}(1000 + 1125 + 1500 + 1000) \quad \text{(Standard five point formula)}$$

$$= 1187.5$$

$$u_4 = \frac{1}{4}[u_1 + 2000 + u_7 + u_5] \quad \text{(Standard five point formula)}$$

$$= \frac{1}{4}[1125 + 2000 + 1125 + 1500] \quad (\because u_1 = u_7)$$

$$= 1437.5$$

$\therefore$ initial values are $u_1 = 1125$

$u_2 = 1187.5$

$u_4 = 1437.5$

$u_5 = 1500$

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(iii) $u(x,0) = 2x$.

We carry out the iteration process using Liebmann's method with standard five point formula

$$u_1^{(n+1)} = \frac{1}{4}\left[1500 + 500 + u_4^{(n)} + u_2^{(n)}\right]$$

$$u_2^{(n+1)} = \frac{1}{4}\left[1000 + 2u_1^{(n+1)} + u_5^{(n)}\right]$$

$$u_4^{(n+1)} = \frac{1}{4}\left[2000 + 2u_1^{(n+1)} + u_5^{(n)}\right]$$

$$u_5^{(n+1)} = \frac{1}{4}\left[2u_4^{(n+1)} + 2u_2^{(n+1)}\right]$$

iteration	u_1	u_2	u_4	u_5
1	1031.25	1140.62	1390.62	1265.62
2	1007.8	1070.31	1320.31	1195.31
3	972.65	1035.15	1285.15	1160.15
4	955.07	1017.57	1267.57	1142.57
5	946.28	1008.78	1258.78	1133.78
6	941.89	1004.39	1254.39	1129.39
7	939.69	1002.19	1252.19	1127.19
8	938.59	1001.09	1251.09	1126.09
9	938.04	1000.54	1250.54	1125.54
10	937.77	1000.27	1250.27	1125.27
11	937.63	1000.13	1250.13	1125.13.

$\therefore u_1 = u_3 = u_7 = u_9 = 937.63$,

$u_2 = u_8 = 1000.13$

$u_4 = u_6 = 1250.13$, $u_5 = 1125.13$.

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5) Solve $u_{xx}+u_{yy}=0$ $(\nabla^2 u=0)$ in the square region bounded by the coordinate axes and the lines $x=4$ and $y=4$ with the boundary conditions given by

$u(0,y)=0$ for $0\le y\le 4$

$u(4,y)=12+y$ for $0\le y\le 4$

$u(x,0)=3x$ for $0\le x\le 4$

$u(x,4)=x^2$ for $0\le x\le 4$.

Also employ Liebmann's method to compute the first ~~Second~~ iterative values of $u(x,y)$.

Soln: We shall divide the square region into 16 squares of side one unit $(h=1)$ 16

we shall derive the values of $u(x,y)$ on the boundary from given equations. Also $x=ih$, $y=jk$ or $y=jh$ since $(h=k)$

(i) $u(0,y)=u(0,jh)=u(0,j)=0$ for $j=0,1,2,3,\ldots$

$\Rightarrow u(0,0)=0$, $u(0,1)=0$, $u(0,2)=0$, $u(0,3)=0$, $u(0,4)=0$

i.e $u_{0,0}=0=u_{0,1}=u_{0,2}=u_{0,3}=u_{0,4}$

(ii) $u(4,y)=12+y$

$\Rightarrow u(4,jh)=12+jh$ $(\because h=1)$

$\Rightarrow u(4,j)=12+j$ for $j=0,1,2,3,\ldots$

i.e $u(4,0)=12$ $u(4,4)=16$.

$u(4,1)=13$

$u(4,2)=14$

$u(4,3)=15$

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(iii) $u(x,0) = 3x$

$\Rightarrow u(ih,0) = 3(ih)$ $(\because h=1)$

$u(i,0) = 3i$ for $i = 0,1,2,..$

$u(0,0) = 0$

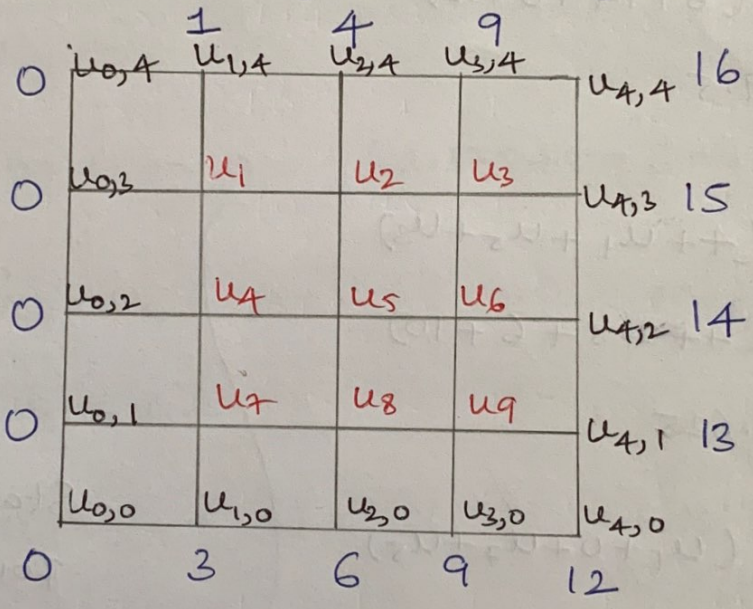
$u(1,0) = 3$, $u(2,0) = 6$, $u(3,0) = 9$, $u(4,0) = 12$

(iv) $u(x,4) = x^2$

$\Rightarrow u(ih,4) = (ih)^2$ $(\because h=1)$

$u(i,4) = i^2$

$\therefore u(0,4) = 0$, $u(1,4) = 1$, $u(2,4) = 4$, $u(3,4) = 9$, $u(4,4) = 16$



To calculate the initial approximation, we take

$u_5 = \frac{1}{4}(4+0+6+14) \Rightarrow u_5 = 6$ (Standard five point formula)

$u_1 = \frac{1}{4}(0+4+0+u_5)$ (Diagonal five point formula)

$= \frac{1}{4}(0+4+0+6) =$

$u_1 = 2.5$

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$u_3 = \frac{1}{4}(4+16+u_5+14)$ (Diagonal five point formula)

$= \frac{1}{4}(4+16+6+14)$

$u_3 = 10$

$u_7 = \frac{1}{4}(0+u_5+0+6)$ (Diagonal five point formula)

$= \frac{1}{4}(0+6+0+6)$

$u_7 = 3$

$u_9 = \frac{1}{4}(u_5+14+6+12)$ (Diagonal five point formula)

$= \frac{1}{4}(6+14+6+12)$

$u_9 = 9.5$

$u_2 = \frac{1}{4}(4+u_1+u_5+u_3)$

$= \frac{1}{4}(4+2.5+6+10)$

$u_2 = 5.625$

$u_4 = \frac{1}{4}(u_1+0+u_7+u_5)$ Standard five point formula.

$= \frac{1}{4}(2.5+0+3+6)$

$= 2.875$

$u_6 = \frac{1}{4}(u_3+u_5+u_9+14)$

$= \frac{1}{4}(10+6+9.5+14)$

$= 9.875$

$u_8 = \frac{1}{4}(u_5+u_7+6+u_9)$

$= \frac{1}{4}(6+3+6+9.5)$

$= 6.125$

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$\therefore$ initial values are $u_1 = 2.5, u_2 = 5.625, u_3 = 10, u_4 = 2.875$

$u_5 = 6, u_6 = 9.875, u_7 = 3, u_8 = 6.125, u_9 = 9.5$

Using these values as initial approximation to Liebmann's method, we get

(Standard five point formula to all)

$$u_1 = \frac{1}{4}(1+0+u_2+u_4) = \frac{1}{4}(1+0+5.625+2.875)$$

$$u_1 = 2.375$$

$$u_2 = \frac{1}{4}(4+u_1+u_3+u_5) = \frac{1}{4}(4+2.375+10+6) = 5.5937$$

$$u_3 = \frac{1}{4}(9+u_2+u_6+15) = \frac{1}{4}(9+5.5937+9.875+15)$$

$$u_3 = 9.8671$$

$$u_4 = \frac{1}{4}(u_1+0+u_7+u_5) = \frac{1}{4}(2.375+0+3+6) = 2.8437$$

$$u_5 = \frac{1}{4}(u_2+u_4+u_6+u_8) = \frac{1}{4}(5.5937+2.8437+9.875+6.125)$$

$$u_5 = 6.1093$$

$$u_6 = \frac{1}{4}(u_3+u_5+u_9+14) = \frac{1}{4}(9.8671+6.1093+9.5+14)$$

$$u_6 = 9.8691$$

$$u_7 = \frac{1}{4}(u_4+0+3+u_8) = \frac{1}{4}(2.8437+0+3+6.125)$$

$$u_7 = 2.9921$$

$$u_8 = \frac{1}{4}(u_5+u_7+6+u_9) = \frac{1}{4}(6.1093+2.9921+6+9.5)$$

$$u_8 = 6.1503$$

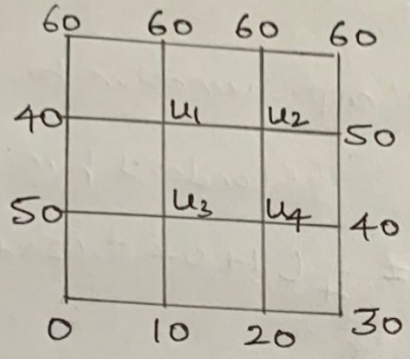
$$u_9 = \frac{1}{4}(u_6+u_8+9+13) = \frac{1}{4}(9.8691+6.1503+9+13)$$

$$u_9 = 9.5048$$

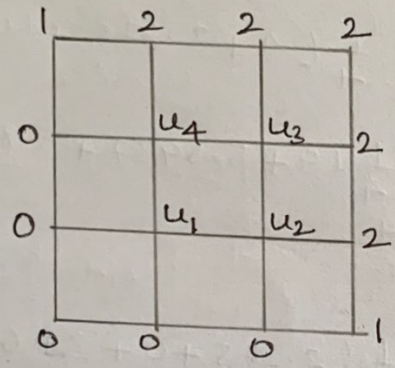
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Solve the following Laplace equation with boundary conditions as indicated in the figure

(1)



(2)



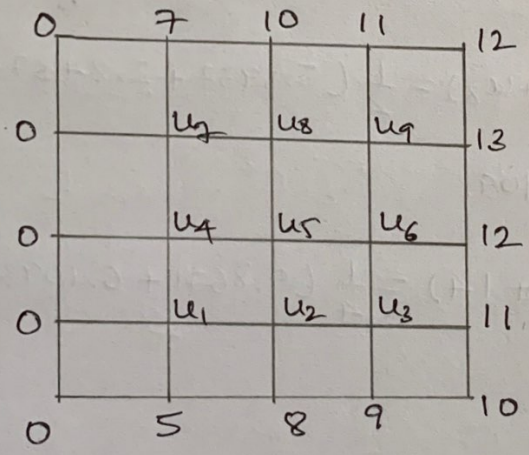
Answer:

~~Hint~~

$u_1 = 0.5,\ u_2 = 1,\ u_3 = 1.5$

$u_4 = 1$

(3)



One iteration only

Answer

$u_1 = 4.0312$ $u_6 = 9.7261$

$u_2 = 7.2266$ $u_7 = 4.7598$

$u_3 = 9.2598$ $u_8 = 8.101$

$u_4 = 3.9766$ $u_9 = 10.48$

$u_5 = 7.2696$

(4) Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for $0 < x < 1$, $0 < y < 1$, given $u(x,0) = u(0,y) = 0$

$u(x,1) = 6x,\ 0 < x < 1$

$u(1,y) = 3y,\ 0 < y < 1.$

Divide the region into 9 square mesh $(h = 1/3)$

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